

Sophisticated Agent VFI: Required Changes to the Appendix

1 Current Paper (Naive Agent)

The current paper defines a single choice-specific value function with $\beta\delta$ in the continuation:

$$\bar{v}_{ijt}(\mathbf{d}_{it}, \mathbf{x}_{it}) = \bar{u}_{ijt}(\mathbf{d}_{it}) + \beta\delta \int_{\mathcal{X}} \int_{\mathbb{R}^J} v(\mathbf{x}_{i,t+1}, \boldsymbol{\epsilon}_{i,t+1}) dF_{\boldsymbol{\epsilon}} dF_{\mathbf{x}} \quad (1)$$

The inner integral (integrating out $\boldsymbol{\epsilon}$) uses the fact that the agent both **chooses** and **experiences** the same $\bar{v}_{ijt}$, giving:

$$\int_{\mathbb{R}^J} \max_j \{\bar{v}_{ijt} + \epsilon_{ijt}\} dF_{\boldsymbol{\epsilon}} = \log \left(\sum_{j' \in \mathcal{J}} \exp \{\bar{v}_{ij't}\} \right) \quad (2)$$

And the VFI algorithm stores:

$$V_{\text{choice}}[j, a_h, \mathbf{p}_m] = u + \beta\delta \cdot EV \quad (3)$$

$$V_{\text{next}}[a_h, \mathbf{p}_m] = \log \left(\sum_{j \in \mathcal{J}} \exp(V_{\text{choice}}[j, a_h, \mathbf{p}_m]) \right) \quad (4)$$

2 Sophisticated Agent: What Changes

2.1 Two Choice-Specific Value Functions

A sophisticated agent knows their future self will also be present-biased. This means the agent **chooses** according to one value (decision utility, with $\beta\delta$) but **experiences** another (experienced utility, with δ). We need two objects instead of one:

$$\bar{v}_{ijt}^d(\mathbf{d}_{it}, \mathbf{x}_{it}) = \bar{u}_{ijt}(\mathbf{d}_{it}) + \beta\delta \int_{\mathcal{X}} V(\mathbf{x}_{i,t+1}) dF_{\mathbf{x}} \quad (5)$$

$$\bar{w}_{ijt}(\mathbf{d}_{it}, \mathbf{x}_{it}) = \bar{u}_{ijt}(\mathbf{d}_{it}) + \delta \int_{\mathcal{X}} V(\mathbf{x}_{i,t+1}) dF_{\mathbf{x}} \quad (6)$$

where $V(\mathbf{x})$ is the integrated (ex-ante) value function that we iterate on.

- $\bar{v}_{ijt}^d$: **Decision utility**. What the agent uses to choose (includes present bias β).
- $\bar{w}_{ijt}$: **Experienced utility**. What actually accrues from choosing j (no β on flow utility, standard δ discounting).

Note the key relationship: they share the same EV term, so

$$\bar{v}_{ijt}^d = \bar{u}_{ijt} + \beta\delta \cdot EV, \quad \bar{w}_{ijt} = \bar{u}_{ijt} + \delta \cdot EV. \quad (7)$$

2.2 The Inner Integral Changes

The agent draws shocks ϵ_{ijt} and chooses $j^* = \arg \max_j \left\{ \bar{v}_{ijt}^d + \epsilon_{ijt} \right\}$. But their realized payoff is $\bar{w}_{ij^*t} + \epsilon_{ij^*t}$, **not** $\bar{v}_{ij^*t}^d + \epsilon_{ij^*t}$.

The continuation value is therefore:

$$V(\mathbf{x}_{it}) = \mathbb{E}_\epsilon [\bar{w}_{ij^*t} + \epsilon_{ij^*t}] \quad (8)$$

where $j^* = \arg \max_j \left\{ \bar{v}_{ijt}^d + \epsilon_{ijt} \right\}$.

Since the choice is made according to $\bar{v}^d$ but the payoff is $\bar{w}$, the logsumexp identity (2) **no longer applies**. That identity requires the same vector to appear in both the arg max and the payoff.

Instead, we compute this expectation in two pieces.

Piece 1: Expected experienced utility. The probability of choosing j is:

$$p_j(\mathbf{x}_{it}) = \frac{\exp\left(\bar{v}_{ijt}^d\right)}{\sum_{j' \in \mathcal{J}} \exp\left(\bar{v}_{ij't}^d\right)} \quad (9)$$

The expected experienced utility (deterministic part) is:

$$\mathbb{E}_\epsilon [\bar{w}_{ij^*t}] = \sum_{j \in \mathcal{J}} p_j \cdot \bar{w}_{ijt} \quad (10)$$

Piece 2: Expected shock conditional on being chosen. Under the Type I Extreme Value assumption, the conditional expected value of the shock for the chosen alternative is:

$$\mathbb{E} [\epsilon_{ij^*t} \mid j^* = j] = -\log(p_j) \quad (11)$$

Intuitively, rare choices (low p_j) only occur when ϵ_j is unusually large, so the conditional expectation is higher. Averaging over all alternatives:

$$\mathbb{E}_\epsilon [\epsilon_{ij^*t}] = \sum_{j \in \mathcal{J}} p_j \cdot \mathbb{E}[\epsilon_{ijt} \mid j^* = j] = \sum_{j \in \mathcal{J}} p_j \cdot (-\log p_j) = H(\mathbf{p}) \quad (12)$$

where $H(\mathbf{p}) = -\sum_j p_j \log p_j$ is the entropy of the choice probabilities.

Combined. Substituting into (8):

$$\boxed{V(\mathbf{x}_{it}) = \sum_{j \in \mathcal{J}} p_j(\mathbf{x}_{it}) \cdot \bar{w}_{ijt}(\mathbf{x}_{it}) + H(\mathbf{p}(\mathbf{x}_{it}))} \quad (13)$$

This replaces the logsumexp aggregation from the naive case.

2.3 Why Logsumexp is a Special Case

When $\beta = 1$: $\bar{v}_{ijt}^d = \bar{w}_{ijt}$ (both equal $\bar{u} + \delta \cdot EV$). Then $p_j = \text{softmax}(\bar{w})_j$ and:

$$\begin{aligned}
\sum_j p_j \cdot \bar{w}_j + H(\mathbf{p}) &= \sum_j p_j \cdot \bar{w}_j - \sum_j p_j \log p_j \\
&= \sum_j p_j \cdot \bar{w}_j - \sum_j p_j \left(\bar{w}_j - \log \sum_{j'} e^{\bar{w}_{j'}} \right) \\
&= \sum_j p_j \cdot \log \sum_{j'} e^{\bar{w}_{j'}} \\
&= \log \sum_{j'} e^{\bar{w}_{j'}} \\
&= \text{logsumexp}(\bar{w})
\end{aligned} \tag{14}$$

So the sophisticated aggregation reduces to exactly the naive logsumexp when $\beta = 1$.

2.4 Conditional Choice Probabilities

The CCPs have the same softmax form as before, but use $\bar{v}^d$ (decision utility):

$$p_{ijt}(\mathbf{d}_{it} \mid \mathbf{x}_{it}) = \frac{\exp \left[\bar{v}_{ijt}^d(\mathbf{d}_{it}, \mathbf{x}_{it}) \right]}{\sum_{j' \in \mathcal{J}} \exp \left[\bar{v}_{ij't}^d(\mathbf{d}'_{it}, \mathbf{x}_{it}) \right]} \tag{15}$$

The only difference from the naive case is that V inside $\bar{v}^d$ is now the *sophisticated* value function (equation (13)), not the logsumexp value function.

2.5 VFI Algorithm Changes

The storage step (equation (3)) becomes two arrays:

$$V_d[j, a_h, \mathbf{p}_m] = u + \beta \delta \cdot EV \quad (\text{decision utility}) \tag{16}$$

$$W[j, a_h, \mathbf{p}_m] = u + \delta \cdot EV \quad (\text{experienced utility}) \tag{17}$$

The aggregation step (equation (4)) becomes:

$$p_j = \frac{\exp(V_d[j, a_h, \mathbf{p}_m])}{\sum_{j'} \exp(V_d[j', a_h, \mathbf{p}_m])} \tag{18}$$

$$V_{\text{next}}[a_h, \mathbf{p}_m] = \sum_{j \in \mathcal{J}} p_j \cdot W[j, a_h, \mathbf{p}_m] + H(\mathbf{p}) \tag{19}$$

2.6 Contraction Mapping

The sophisticated operator T is still a contraction with modulus δ . To see this, consider a uniform shift $V \rightarrow V + c$:

- $EV \rightarrow EV + c$
- $V_d \rightarrow V_d + \beta \delta c$ (uniform shift, so softmax probabilities $\mathbf{p}$ are unchanged)

- $W \rightarrow W + \delta c$ (uniform shift)
- $T(V + c) = \sum_j p_j \cdot (W_j + \delta c) + H(\mathbf{p}) = T(V) + \delta c$

Since $\delta < 1$, the operator is a contraction and VFI converges to the unique fixed point.

3 Summary of Changes to the Appendix

Section	Current (Naive)	Sophisticated
Choice-specific value	One: $\bar{v} = \bar{u} + \beta\delta \cdot EV$	Two: $\bar{v}^d = \bar{u} + \beta\delta \cdot EV$, $\bar{w} = \bar{u} + \delta \cdot EV$
Inner integral	$\text{logsumexp}(\bar{v})$	$\sum p_j \bar{w}_j + H(\mathbf{p})$, $p = \text{softmax}(\bar{v}^d)$
CCPs	$\text{softmax}(\bar{v})$	$\text{softmax}(\bar{v}^d)$ (same form, different V)
VFI storage	$V_{\text{choice}} = u + \beta\delta \cdot EV$	$V_d = u + \beta\delta \cdot EV$ $W = u + \delta \cdot EV$
VFI aggregation	$\text{logsumexp}(V_{\text{choice}})$	$\sum \text{softmax}(V_d) \cdot W + H$
Post-convergence	$V_{\text{decision}} = (1 - \beta)U + \beta V_{\text{choice}}$	$V_{\text{decision}} = V_d$ (direct)
Contraction rate	δ	δ (same)
Likelihood	No change	No change

4 Why Does W Use δ and Not $\beta\delta$?

A natural question: why does the experienced utility W discount the continuation with δ rather than $\beta\delta$? The answer is that β is **not** a per-period discount factor. It is a one-time wedge between “right now” and “everything after right now.”

4.1 The Quasi-Hyperbolic Discount Sequence

From the perspective of the self at time t , the discount weights on future periods are:

$$\underbrace{1}_{\text{period } t}, \quad \underbrace{\beta\delta}_{\text{period } t+1}, \quad \underbrace{\beta\delta^2}_{\text{period } t+2}, \quad \underbrace{\beta\delta^3}_{\text{period } t+3}, \quad \dots$$

Now look at the discount *between consecutive future periods*:

$$\begin{aligned} \text{From } t \text{ to } t+1 : \quad & \frac{\beta\delta}{1} = \beta\delta \\ \text{From } t+1 \text{ to } t+2 : \quad & \frac{\beta\delta^2}{\beta\delta} = \delta \\ \text{From } t+2 \text{ to } t+3 : \quad & \frac{\beta\delta^3}{\beta\delta^2} = \delta \end{aligned}$$

The β cancels between any two consecutive *future* periods. It only appears as a wedge between the present (t) and the future ($t+1$ onward). This means the agent can factor the future stream as:

$$\underbrace{U_t}_{\text{now}} + \beta\delta \cdot \underbrace{\left[U_{t+1} + \delta \cdot U_{t+2} + \delta^2 \cdot U_{t+3} + \dots \right]}_{\text{the future, discounted at rate } \delta} \quad (20)$$

The term in brackets is the δ -discounted future stream. This is W — the experienced value. The β just scales the entire bracketed term down from the current self’s perception.

4.2 So What Exactly is W ?

$W[j] = U[j] + \delta \cdot EV$ is the actual δ -discounted value of choosing alternative j . It answers the question: “if I choose j today, what is the true present value of the resulting payoff stream (discounted at the patient rate δ)?”

$V_d[j] = U[j] + \beta\delta \cdot EV$ is the same thing but *perceived* through the lens of present bias. It answers: “how valuable does choosing j *feel* to me right now, given that I over-discount the future by β ?”

The β doesn’t destroy real value — it distorts the agent’s *perception* of value. Think of it as tinted glasses: the future looks dimmer (less valuable) while you’re wearing them, so you make impatient choices, but the actual payoffs are still there at their full δ -discounted value.

4.3 This is the Same in Both Naive and Sophisticated

Both the naive and sophisticated agent have the **exact same** building blocks:

$$\begin{aligned} W[j] &= U[j] + \delta \cdot EV && \text{(experienced utility — same in both)} \\ V_d[j] &= U[j] + \beta\delta \cdot EV && \text{(decision utility — same in both)} \end{aligned}$$

The discount rates are identical. The δ in W and the $\beta\delta$ in V_d are the same for naive and sophisticated agents. So the question becomes: if the building blocks are the same, what is actually different?

4.4 The Only Difference: The Aggregation Step

The difference is in how the continuation value V is computed from V_d and W .

Naive agent. The naive agent believes their future self will be a patient δ -discounter who chooses according to W . So the future self's choice probabilities are $\text{softmax}(W)$, and the continuation is:

$$V^{\text{naive}} = \sum_j \text{softmax}(W)_j \cdot W[j] + H(\text{softmax}(W)) = \text{logsumexp}(W) \quad (21)$$

This is just logsumexp of W (by the identity in equation (14)).

Sophisticated agent. The sophisticated agent knows their future self will be present-biased and will choose according to V_d . So the future self's choice probabilities are $\text{softmax}(V_d)$, and the continuation is:

$$V^{\text{soph}} = \sum_j \text{softmax}(V_d)_j \cdot W[j] + H(\text{softmax}(V_d)) \quad (22)$$

This is **not** logsumexp of anything, because the probabilities come from V_d but the payoffs come from W — two different vectors.

4.5 Why the Aggregation Matters

When $\beta < 1$, V_d and W differ. The decision utility V_d puts relatively more weight on flow utility U (because the $\beta\delta \cdot EV$ term is shrunk relative to $\delta \cdot EV$). This means $\text{softmax}(V_d)$ tilts toward alternatives with high immediate gratification compared to $\text{softmax}(W)$.

Concretely, consider two alternatives:

- Alternative A: high U (tastes good now), low EV (bad for long-run addiction)
- Alternative B: low U (abstain), high EV (good for long-run health)

The patient probabilities $\text{softmax}(W)$ balance U and $\delta \cdot EV$, giving reasonable weight to B. The impatient probabilities $\text{softmax}(V_d)$ shrink the EV term by β , tilting toward A.

Since A has lower experienced value W than the patient choice would deliver, the sophisticated continuation value is lower:

$$V^{\text{soph}} < V^{\text{naive}}$$

The sophisticated agent correctly predicts that future present bias will destroy some welfare.

4.6 The Feedback Loop

This difference in V feeds back through EV on the next VFI iteration:

1. Different $V \Rightarrow$ different EV
2. Different $EV \Rightarrow$ different V_d and W
3. Different $V_d \Rightarrow$ different choice probabilities $\mathbf{p}$
4. Different $\mathbf{p}$ and $W \Rightarrow$ different V on the next iteration
5. Repeat until convergence

So even though the per-iteration formulas for V_d and W look the same, the *entire value function* is different between naive and sophisticated because the aggregation step propagates through every iteration.

When $\beta = 1$: $V_d = W$, so $\text{softmax}(V_d) = \text{softmax}(W)$, both aggregations give the same V , and the feedback loop produces identical results. The naive and sophisticated models are indistinguishable.